Duty at home

DM7 · An evolution with variable input · 60 min.

We're looking for a function y defined and derivable on $[0;+\infty[$ such that $y'(t) = -y(t) + 2t + 1$ and $y(0) = 3$.

1. Determine two actuals a and b so that $g(t) = at + b$ or a differential equation solution.
2. Deduce the expression of the solution that verifies $y(0) = 3$.
3. Study the variations of this solution and specify its minimum as well as the moment it is reached.
4. Write a Python argument function N, integer strictly positive, which applies the method d-Euler on $[0;2]$ with the step $\frac{2}{N}$ and returns the approximation of $y(2)$. Compare with the exact value for $N = 10$, then $N = 100$.
5. Why is the verification of differential equality in just a few moments not enough to prove that a function is a solution?